

Vernacular Pedagogy App — Content & Feature Plan

SIH 27042 | Hindi → Santhali / Ho / Mundari

This document lists everything the app needs to contain, with sample content, so the frontend/theme design has a real basis to work from.

1. App Entry Flow

Step 1 — Language Selector (first screen, before anything else)

- Big tappable tiles, one per language: Santhali, Ho, Mundari (+ Hindi, for teacher setup)
- Each tile shows the language's own script/name, plus a tap-to-hear audio sample ("This is Santhali" spoken in Santhali) — so non-readers can pick correctly
- Selection sets the language for the **entire app**, not just content — every button, label, and instruction switches too
- Recommended: language is set **per device**, since tablets will likely be assigned per classroom, not per student login

Step 2 — Home Screen

- One large "Translate / Speak" button (core feature, backend by teammate)
- Icon tiles for each module below (icons + one-word caption, no long text)

Two Modes:

Mode	Who	What it shows
Student Mode	Kids	Selected tribal language only, big icons, audio-first
Teacher Mode	Teachers	Tribal language + Hindi side-by-side, plus dashboard/progress tools

2. Content Modules (with sample content)

A. Flashcards

Category decks, each card = image + word (text + audio) in selected language.

Decks to include: Alphabets, Numbers, Animals, Family/Relations, Colors, Body Parts, Classroom Objects, Food, Local Festivals (Sarhul, Karma), Farming Tools

Example card set (Animals):

Image	Word slot
 Cow	[language word] + audio
 Goat	[language word] + audio
 Hen	[language word] + audio
 Elephant	[language word] + audio
 Dog	[language word] + audio
 Cat	[language word] + audio
 Pig	[language word] + audio
 Sheep	[language word] + audio
 Bird/Sparrow	[language word] + audio
 Monkey	[language word] + audio

For class 4-6: add a swipe "I know it / I don't know it" sorting mode.

Example card set (Numbers 1-10):

Image	Word slot
 (one object shown)	[language word for "one"] + audio
 (two objects shown)	[language word for "two"] + audio
 (three objects shown)	[language word for "three"] + audio
 (four objects shown)	[language word for "four"] + audio
 (five objects shown)	[language word for "five"] + audio
 (six objects shown)	[language word for "six"] + audio
 (seven objects shown)	[language word for "seven"] + audio
 (eight objects shown)	[language word for "eight"] + audio
 (nine objects shown)	[language word for "nine"] + audio
 (ten objects shown)	[language word for "ten"] + audio

Each number card pairs the numeral with a matching count of simple objects (e.g., 3 mangoes for "3"), not just the digit alone — this reinforces quantity recognition, not just symbol memorization. Extend the deck to 20 for class 2-3, and introduce tens/multiples of ten for class 4-6.

B. Image Study (Scene Vocabulary)

One large illustrated scene with tappable hotspots. Tap an object → hear + see its name in the selected language.

Scenes to include: Classroom, Market, Farm/Field, River/Forest, Home, Festival gathering

This works even for children who can't read yet — pure audio-visual learning.

C. Worksheets (auto-generated, template-based)

Type 1 — Match the Following (Class 1-2) Image on one side, word-audio bubble on other side; child connects them.

Example set: Cow / Goat / Hen / Elephant / Dog (images ↔ language audio bubbles)

Type 2 — Fill in the Blank (picture clue) (Class 1-3) Picture shown, one letter/sound missing from the word underneath, in the selected language script/transliteration.

Type 3 — Trace & Draw (Class 1) Trace a number, then draw that many objects.

Example: Trace "3" → draw 3 apples

Type 4 — Picture Word Problem (Class 3-4, Numeracy) Image-based story, question asked in selected language + audio.

Example: Image shows 4 mangoes on a tree + 2 falling. Question (in language): "How many mangoes in total?"

Type 5 — Sentence-to-Picture Match (Class 5-6) Short sentence (text + audio, selected language) matched to the correct picture.

Examples: "The child is going to school" → child+bag+school icon; "Mother is cooking" → mother+pot icon; "The farmer is working in the field" → farmer+field icon

Teacher toggle: every worksheet can show Hindi alongside the tribal language for the teacher's own reference/prep — not shown to students by default.

D. Read-Along Storybooks

Illustrated short stories, sentence-by-sentence audio with text highlighting as it plays (karaoke-style).

- Use local folk stories / tribal cultural stories where possible, not generic imported stories

- Fully in selected language; Hindi available as teacher reference only
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E. Numeracy Module

- Counting with visual objects (tap-to-count)
 - Shape and pattern recognition
 - Addition/subtraction shown visually before numerically (e.g., picture of 3 mangoes + 2 mangoes, then the equation)
 - Word problems using locally relevant items (paddy, mahua, forest produce) instead of generic apples/oranges
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F. Rhymes & Songs

Short animated rhymes/songs, audio-first, in the selected language. Strong for early language acquisition, minimal reading required.

G. Speaking Practice Game

Child taps a picture → hears the word → repeats it → gets a simple star/badge reward. Doesn't need perfect speech recognition — approximate matching is enough for engagement.

H. Teacher Dashboard

- Class progress overview (which modules/worksheets completed)
 - Auto-flagged "struggling" words/concepts based on flashcard/worksheet performance
 - Lesson planner: pick a Hindi FLN lesson topic → app shows the matching translated + visual version to teach with
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3. Gamification Layer

- Star/badge per completed module
 - Visual "learning journey map" (game-level-style path) instead of a plain progress bar
 - Daily streak icon (e.g., growing tree/sun) for repeat use
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4. Cultural/Visual Direction (for theme & illustration)

- Tribal motifs, local clothing, local flora/fauna in all illustrations — avoid generic clip-art style
 - Vocabulary and word-problem examples should reflect local life: farming tools, forest produce, festivals, regional food
 - This is the main way to differentiate from other teams' generic-themed submissions
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5. Data Structure Note (for whoever builds content/backend)

Every piece of content (flashcard, worksheet item, story line) should be stored with **all languages as equal fields**, not Hindi-as-default:

```
concept_id: animal_cow
image: cow.png
text: { hindi: "गाय", santhali: "", ho: "", mundari: "" }
audio: { hindi: "hi_gaay.mp3", santhali: "", ho: "", mundari: "" }
```

The app just displays `text[selected_language]` and `audio[selected_language]` everywhere — this makes language-switching and adding new content straightforward, and lets the translation and frontend halves of the team work in parallel using the same content IDs.

6. Content Pool to Prepare (vocabulary/sentence bank)

- **Vocabulary sets:** animals, fruits/vegetables, family members, classroom objects, body parts, colors, numbers 1-100, days/months, weather, festivals, farming tools
 - **Sentence patterns:** simple subject-verb-object, question forms (what/who/where), daily routine sentences
 - **Numeracy progression:** counting → addition/subtraction with objects → simple word problems → measurement (bigger/smaller, more/less)
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7. Suggested Build Priority (for prototype/demo)

1. Language selector + Flashcards + Image Study (strongest visual demo)
2. One worksheet type (Match the Following) with auto-generation
3. One read-along story
4. Basic teacher dashboard
5. Gamification polish